

Oceans

The oceans form by far the largest continuous habitat on earth, and they were almost certainly the environment in which life first evolved. The underwater landscape is made up of mountains and volcanoes, cliffs, deep valleys, and vast, flat plains, many of them far larger than any found on land. The oceans are so immense—they cover more than three-quarters of the earth's surface—and difficult to explore that scientific knowledge of ocean wildlife lags behind that of life on land. However, research has shown that life is found at all levels, from the sunlit surface to the deepest trenches over 7 miles (11 km) down.

Inshore waters

Some inshore waters are so shallow that if the world's oceans were lowered by just 245 ft (75 m), huge areas of seabed would be exposed. Off Western Europe, for example, the coast would be extended by about 125 miles (200 km), and in parts of Siberia, by more than 435 miles (700 km).

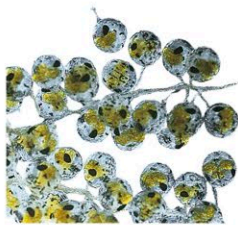
These shallow waters owe their existence to continental shelves—the gently sloping plateaus that flank many of the deep ocean basins.

Continental shelves are a key habitat for sea life, supporting large schools of fish and a diverse collection of other animals, from lobsters and crabs to mollusks and burrowing worms. This wealth of life is possible because, in shallow water, sunlight can reach the seabed, promoting the growth of algae, seagrasses, and countless other organisms that need energy from light to survive. Just like plants on land, these provide animals with a year-round supply of food, as well as with plenty of cover and places to breed.

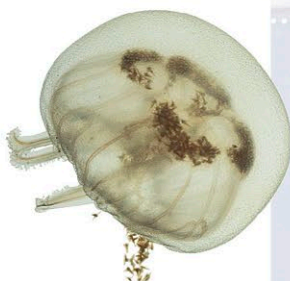
Some inshore animals, such as lobsters and flatfish, spend most of their lives on the seabed, while a few live entirely in midwater or at the surface. Others live in quite different habitats at different stages of their lives. For example, some

larvae develop as part of the surface plankton, then move to midwater or the seabed as adults.

Inshore waters are also visited by animals from the open sea—most often by passive drifters, such as jellyfish, but also by powerful swimmers, such as whales and sharks. Some come to breed, while others arrive by accident and then return to deeper water. Occasionally, the latter get into difficulties: jellyfish often end up on the shore, and whales can become stranded when their navigation systems guide them into shallow water instead of safely out to sea.



PLANKTONIC LIFE near the surface of inshore waters consists partly of sea animals starting their lives as larvae.



JELLYFISH, the largest individual planktonic animals, can swim as well as drift near the surface. Some are over 6½ ft (2 m) across.



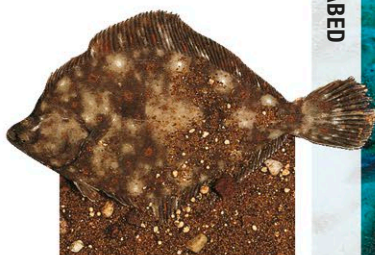
MOST SHARKS live in the relatively shallow water of the continental shelves rather than in the deep sea.



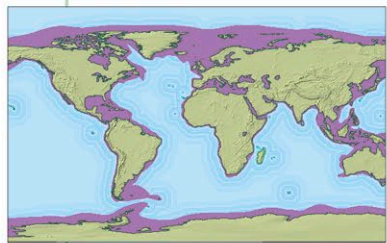
FISH such as cod shed vast numbers of eggs near the seabed. The eggs float gradually upward, and the young fish live near the surface.



LOBSTERS are common seabed scavengers, crawling over the bottom in search of live prey and dead remains.



LIKE MANY FLATFISH, the European plaice lives on the seabed. It is camouflaged to match the sediment.



CONTINENTAL SHELVES vary in width from a few miles—in coasts close to deep-sea trenches—to over 620 miles (1,000 km).

SURFACE

MIDWATER

SEABED